Sherif Morsi

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Summary

Computer Science student building deep learning for finance: **PyTorch** models trained from scratch, **portfolio construction** through a differentiable optimizer, and **LLM agents** that act inside signed bounds, each tested out of sample.

Education

Concordia University Montreal, QC, Canada
Bachelor of Computer Science *Expected May 2028*

- **Availability:** Winter 2027 (Jan to Apr 2027) for 4, 8, 12, or 16 months, anywhere in Canada
- **Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Artificial Intelligence, Machine Learning, Linear Algebra, Multivariable Calculus
- **Activities:** Member, QUARCC (Quantitative Research and Competitions Club)

Experience

AI & Machine Learning Intern Aug 2026 - Sep 2026
Nokia *Cairo, Egypt*

- Trained and evaluated Python and **PyTorch** models on wireless network telemetry to forecast 5G **energy demand**
- Developed **data and ML pipelines** for 5G energy telemetry, to support automated network management and optimization
- Trained in **agentic AI** for intent-based 5G network slicing (Amazon Bedrock); presented findings to cross-functional data science and engineering teams and led the project proposal and requirements review

Credit Analyst Job Shadow Aug 2025 - Sep 2025
Qatar National Bank (QNB) *Cairo, Egypt*

- Built and documented **Python** utilities and **SQL** queries that preprocess and validate SME financial records for the bank's credit-review process, improving data quality and reliability

Projects

Deep Portfolio Constructor (Deep Learning, Quant) | *Python, PyTorch, cvxpylayers* | **Project Page** 2026

- Market-neutral long-short portfolio of 99 large caps with **PyTorch at every layer**: a cross-sectional **transformer** return model, a transformer covariance model, and **end-to-end training through a differentiable optimizer**
- Neutrality constraints raised the share of risk from intended stock bets from **17% to 61%**; end-to-end training gave the best net Sharpe (**0.58 vs −0.03**, 225-month walk-forward from 2008, t = 1.70, not significant)
- Diagnosed a **beta leak** (realized beta 0.19 vs a 0.05 limit; hedging it cut Sharpe from 0.64 to 0.40) and a negative result (the fully deep model scored 0.15); **89 tests**, incl. gradient checks through the optimizer

Time-Series Foundation Model (Transformers) | *Python, PyTorch, GluonTS* | **GitHub** | **Project Page** 2026

- Pretrained a **10.1M-parameter** decoder-only transformer from scratch on **443,258** series across 77 datasets; on the frozen test (97 configurations, one seed), CRPS was **0.688**× AutoARIMA, 95% CI [0.624, 0.757]
- Wrote RMSNorm, **rotary embeddings**, SwiGLU, a KV cache and a non-crossing quantile head without a model library; trained in **mixed precision** on one T4 GPU
- A leakage audit found benchmark test data in a public corpus: removed **801 of 447,229** series (642 exact or rescaled copies); 170 tests in CI

World Shock Lab (Econometrics, ML) | *Python, PyTorch, DuckDB, TypeScript* | **GitHub** | **Live Site** 2026

- Estimated how Fed, Bank of Canada, and OPEC surprises move **53** assets (a 10 bp Fed surprise moves the TSX **−1.10%**, 90% band −1.59 to −0.61), replicated **11 of 13** published findings, and publishes forecasts before each decision
- **Pre-registered** test of machine learning on 168 held-out announcements: a neural net did not beat the linear model on the mean (gap −0.003, 90% CI includes zero); quantile boosting won only on 5% tail loss

Aurvane (Agent Security, Delegation) | *Python, Ed25519, MCP, JavaScript* | **GitHub** | **Live Demo** 2026

- Trust layer between LLM agents and their tools: **Ed25519-signed mandates**, four checks per action, Allow/Deny approvals, narrower-only sub-agent mandates, hash-chained log; Python SDK and **MCP server**; 46 tests
- Against a simulated agent that obeys every **prompt injection** (60 paired tasks), attacks that landed fell from **30/30 to 7/30** and money lost from 12,135 to **135 CAD**, with 0 false alarms

Technical Skills

Machine Learning: **PyTorch**, **Transformers**, Differentiable Optimization, Time-Series Forecasting, **scikit-learn**, TensorFlow
Quant / Finance: **Portfolio Optimization** (cvxpy), Factor and Risk Models, **Walk-Forward Backtesting**, Econometrics
AI Agents: **LLM Agents**, **Multi-Agent Systems**, **Tool Calling**, **MCP**, Prompt-Injection Defense, **QLoRA**, LLM Evals
Languages: **Python**, **SQL**, TypeScript, JavaScript, C++, Java, C, R, Bash
Data / Tools: **Pandas**, **NumPy**, **FastAPI**, DuckDB, **Git**, **GitHub Actions (CI/CD)**, **pytest**, Playwright, Docker, Streamlit

Certifications

Anthropic: Building with the Claude API, Claude Code 101, Claude 101 | **IBM:** Machine Learning with Python